



Mutagenesis Approaches in Plants and their Applications

M. Sridhar Reddy ^a and Osman Basha Pinjari ^{b*}

^a *Department of Environmental Science, Yogi Vemana University, Kadapa-516005, Andhra Pradesh, India.*

^b *Department of Genetics and Genomics, Yogi Vemana University, Kadapa-516005, Andhra Pradesh, India.*

Authors' contributions

This work was carried out in collaboration between both authors. Both authors designed the study, managed the literature searches and wrote the review article. Both authors read and approved the final manuscript.

Article Information

DOI: 10.56557/AJAAS/2023/v6i135

Review Article

Received: 04/01/2023

Accepted: 06/03/2023

Published: 11/03/2023

ABSTRACT

Mutagenesis refers to the process of generation of mutations in DNA molecules, including chromosomal rearrangements and alterations in DNA sequence. The incidence of natural or spontaneous mutation frequency in nature is very low, and it is not sufficient for plant breeding programs to develop new varieties. Mutants were also used for gene function discovery which controls the particular phenotype. Genetic diversity was induced in plants on a good scale through induced mutagenesis programmes, and this helped to improve crops all over the world. Significant efforts have been made to develop induced mutagenized resources using physical (Fast-neutron, X-ray, and γ -irradiation), chemical (DEB, EMS, and MNU), and biological (T-DNA and Transposons) agents. However, mutants developed through biological agents or genetically modified organisms (GMO) face severe regulatory limitations imposed by national and international agencies to release as varieties. The most popular chemical and physical mutagenesis techniques used by breeders are EMS and gamma rays mutagens. Estimating the appropriate mutagen dose and mutagen exposure time are essential factors to develop mutagenized resources. The dosage and duration of a mutagenic treatment are determined by the species, physiological state of the species, and material handling prior to and after the mutagenic treatment. Neutrons and X-rays are powerful mutagens that cause significant chromosomal damage and have been used to create large chromosomal deletions.

*Corresponding author: E-mail: osmanbasha@yahoo.co.in;

Keywords: *Mutagenesis; chemical mutagen; physical mutagen; forward genetics.*

1. INTRODUCTION

Agriculture has been the mainstay of the global economy since the beginning of human civilization, with most people relying on it for food, shelter, and clothing. The current agricultural sector produces enough food grains, pulses, fiber, oilseeds and other commodities to feed the world's population while also providing raw resources for manufacturing other products. Following World War II, the progression of extensive scientific understanding, development of new high-yielding varieties, inclusion of automation methods in agriculture, and increased use of chemical pesticides and fertilizers resulted in increased yields. The widespread use of chemical pesticides and fertilizers has had a negative impact on ecology and agricultural employment [1,2]. Agronomic production has nearly tripled in the last 50 years as a result of cutting-edge technology and the Green Revolution. Modern farming is primarily based on plant varieties that perform well despite the excessive use of fertilizer and pesticides. Presently the usage of these inputs is at its highest since these varieties do not give higher yields under low application of fertilizers and pesticides. Nevertheless, on the other worst side, the accessibility of these fertilizers and pesticides will not remain endless and on the other side, the condition is frightening because of the deprivation of natural resources. The upsurge in population growth together with urbanization and industrial development has directed to the reduction in per capita availability of land for farming, as a consequence, worldwide assets are in decline with the increasing human population.

According to new United Nations estimates, the world population will reach 8.6 billion in 2030, 9.8 billion in 2050, and 11.2 billion in 2100 (<https://www.un.org/en/desa/world-population>). This uncontrolled population will create a massive demand for food and raw materials to meet the basic needs of the human population. Crop productivity is generally dropping as a result of soil depletion, which shows up as lower nutrition, reduced water holding capacity, and toxicity. To overcome these problems, mutagenesis approach is a suitable solution in

agriculture in order to develop new varieties and understanding the molecular mechanism of disease resistance, yielding parameters and deal inventive and economically-viable chances for farmers [3].

2. CONVENTIONAL PLANT BREEDING

The concept of "plant breeding" is as old as agriculture itself, with people domesticating crops such as barley and emmer wheat around 10,000 years ago [4]. Plant breeding as an area of science can be traced back to Mendel's experiments on the inheritance of genetic traits such as plant height, seed color, seed shape, pod color, pod shape, and flower position in the early 1860s. Plant breeding is nothing but science that targets to bring improved varieties to agriculturalists through choice/selection in hereditarily variable plants [5]. In general, plant breeding involves the skilful hybridization of plants within the same or closely related species and picking the best appropriate plants based on phenotype.

Plant breeding is synonymous with 'crop improvement' and is basically the procedure of crossing the finest parent species, identifying and recovering the finest offspring over the parents and making them resistance to pests and diseases, improving quality, higher yields, and other improved desirable changes, finally to more useful for human consumption. The progression of plant breeding includes "germplasm collections, identification of superior phenotypes and their hybridization to develop improved cultivars". Besides, the pre-eminent plants are selected from the diversified population; they are then stabilized by sustaining the selected parameter for several years. These newly developed better-quality varieties with superior characters are then verified in diverse climates for their appropriateness, reproduced, and circulated among farmers and the procedure may take 12 to 15 years. In this regard, the "Veery variety of wheat is a good instance of plant breeding due to its tolerance to heat, cold and drought and it was developed through 3,170 crosses involving 51 crop parents collected from 21 countries and released the line in 35 countries" (<https://www.croptrust.org/news-events/news/planet-wheat/>).

3. MUTAGENESIS AN EXCELLENT ROUTE FOR NEW CROP VARIETIES DEVELOPMENT

Induced mutagenesis is a method of exposure of mutagen viz., may be chemical or physical or biological agents to living cells/organisms to enhance the genetic variability at a higher frequency, in comparison to natural mutations. The goal of mutagenesis is to develop raw materials (mutants) for genetic improvement of crop plants as well as to study functional genomics [6]. Mutagenesis research programs have generated a huge range of genetic variability and have played a substantial role in plant breeding programs all over the world. During mutagenesis, introduction of altered alleles of genes in mutants leads to the enhancement of genetic variation [7]. The enhanced genetic variation with heritable changes in genetic material is an excellent aspect and breeders used these developed materials to produce new varieties [8]. In plant mutagenesis, the creation of new genetic variation in inbred varieties coupled with high-resolution reverse and forward genetics screening methods allowed breeders to choose traits of interest and this has been giving a unique possibility to identify novel traits when was very difficult for a few decades ago. With the rapid gathering of whole genome sequence information, the development of advanced molecular markers technology and continuous reduction of the costs related to these techniques had helped breeders to reduce time and money for developing new varieties [9].

4. TYPES OF MUTATIONS

The modifications that happened in the DNA sequences are called mutations and these mutations were valuable for crop improvement. The mutations were classified based on how the sequence alteration affects the gene function and its protein function. In general, the mutations may be single bases modifications, single-base deletions, insertions and more than single-base pair deletions. One base is replaced by another base pair (transitions and transversions) in point mutations. In transition one purine (A or G) was replaced by another purine or pyrimidine (C or T) was replaced by another pyrimidine. Transversions were defined as purine being replaced by a pyrimidine or vice-versa [10].

Any modification in point mutations that doesn't have any effect if any modifications happened

and it has no effect in the amino acid sequence due to wobble base is called a "Silent Mutation" and these mutations cannot be detected by phenotypic change. Any modification that happened in a single nucleotide that leads to amino acid change was referred to as missense mutations and these mutations may or may not lead to detectable phenotypic changes. Another class of point mutation is nonsense mutations and these mutations were described by the introduction of the stop codon instead of the amino acid codon which results in premature termination of the translation and generally, these mutations were inactive. The addition of one or more than one base in the DNA sequence is called insertion and deletion of one or more than one base from the sequence is called as deletion and these mutations generally are referred to as "Indels" [11]. Indels result in a frameshift mutation and the general consequence is the loss of function of a gene. Sometimes three base pair insertions or deletions (or multiples) are more inoffensive as the reading frame tends to be preserved. Through chemical, physical, and insertional mutagenesis approaches, significant efforts have been made to produce large-scale mutagenized populations. Chemical agents like MNU, DEB, EMS, or physical agents like fast-neutron, gamma rays, UV radiation, X-ray irradiation, etc. as well as insertional mutagens like transposable elements or T-DNA were used to create mutagenized populations for the development of novel alleles [12-14]. Physical and chemical mutagenesis agents both cause plant mutations; moreover, their mechanisms of action and efficiency vary. Several research laboratories are actively working on the generation and high throughput screening of mutagenized populations of crops using reverse genetic approaches.

4.1 Chemical Mutagens

Chemical agents are widely known because they are simple to use, do not involve sophisticated equipment, and cause higher mutation incidences. Diverse mutations, including single base pair alterations, deletions, translocations, and point mutations can be induced by chemical mutagens, base analogs, intercalating agents, base modifiers, etc. are examples of chemical agents. 5-bromouracil (BU) is one of the classical examples of a base analog which incorporated into DNA during replication as nucleotide bases and introduces transition or transversion types of mutations [15]. Alkylating agents such as ethyl

Table 1. EMS-induced mutagenized population development in plant species

S. No	Species	Mutagen	Source
1	Maize	EMS	Till et al. [18]
2	Durum wheat	EMS	Slade et al. [19]
3	Rice	EMS	Till et al. [20]
4	Pea	EMS	Triques et al. [21]
5	Soybean	EMS	Cooper et al. [22]
6	Barley	Az	Talame et al. [23]
7	Rapeseed	EMS	Wang et al. [24]
8	Sorghum	EMS	Xin et al. [25]
9	Tomato	EMS	Gady et al. [26]
10	Cabbage	EMS	Himelblau [27]
11	Pea	EMS	Dalmais et al. [28]
12	Oat	EMS	Chawade et al. [29]
14	<i>Brassica rapa</i>	EMS	Stephenson et al. [30]
15	Sunflower	EMS	Sabetta et al. [31]
16	<i>Arabidopsis thaliana</i>	EMS	Jander et al. [32]
17	<i>Chrysanthemum morifolium</i>	EMS	Nasri et al. [33]
18	<i>Capsicum annuum</i>	EMS	Siddique et al. [34]
19	<i>Lotus japonicus</i>	EMS	Perry et al [35]
20	Piel de Sapo melons	EMS	González et al. [36]
21	<i>Vigna unguiculata</i>	EMS	Opoku Gyamfi et al. [37]
22	<i>Antirrhinum majuss</i>	EMS	Heffron et al. [38]
23	<i>Vigna mungo</i>	EMS	Goyal et al. [39]
24	Groundnut		Karaman et al. [40]
25	<i>Dendranthema grandiflora</i>		Latado et al. [41]

methanesulfonate (EMS), methyl methanesulfonate (MMS), and epoxides causes mutations [16]. EMS (ethyl methanesulfonate) chemical agent is most widely used mutagen to induce mutagenesis in crop plants. EMS mutagen alkylated guanine (G) and during replication, thymine residue was added instead of cytosine residue and a 70–99% majority of the GC to AT base mutations happened in the mutagenized population. EMS is most popular mutagen used to develop a mutagenized population for the identification of new alleles (Table 1). Methyl nitrosourea (MNU) and Sodium azide (Az) chemicals were also used to develop a mutagenized population and mode of base pair changes different from EMS. Acridine orange and ethidium bromide are placed in another class of mutagen called "Intercalating agents" which created stretch (intercalated) in gene sequence and help in insertion of an extra base and these result in frameshift mutations [17].

4.2 Physical Mutagens

Historically, a variety of strategies for inducing mutations developed in the middle of the twentieth century to avoid the genetic deterioration of elite crops. The knowledge of induction of mutation by X-rays as mutagens was

developed later in 1927. Roentgen and Marie and Pierre Curie received Nobel Prizes for their discoveries of X-rays and radioactive elements. These discoveries completely changed the mutagenesis programs through induction of mutations in animals and plants. X-ray radiation was used to induce and enhance the mutation rate in *Drosophila* species and noted a 150 times higher lethal mutation rate by Muller [42]. After Muller, another scientist Stadler et al. [43,44] noted a strong correlation of phenotypic variation of barley seedlings and sterility in maize after the treatment of X-rays and radium (Food and Agriculture Organization of the United Nations, 2011). The electromagnetic (EM) spectrum which includes cosmic, gamma (γ) and X-rays have the capability to change nucleotide sequence in the genome after the treatment [45, 46]. Later, more sophisticated techniques such as gamma and neutron radiation were developed and these techniques replaced X-ray radiation in mutagenesis programs [47, 48]. Gamma radiation created point mutations as well as small deletions and fast neutron treatment causes translocations and large deletions. In the gamma radiation facility plants were exposed to irradiation for standardized periods and these facilities were controlled by only sufficiently trained persons and depending on national

statutory requirements. Gamma rays are released from the Cesium-137 (¹³⁷Cs), Cobalt-60 (⁶⁰Co), and to a less extent, Plutonium-239 (²³⁹Pu) radioisotopes. Ultraviolet (UV) light also has the capability to induce mutations [49].

In the early and mid-1990s, the method of implantation of ion beams was starting to gain significance in China and Japan for inducing mutations in crops [50,51]. Ion beams are different from X-rays, Gamma rays, fast neutrons and UV rays. Ion beams are produced by particle accelerators. High-energy particles originating in outer space (Cosmic radiation) have also proved to be mutagenic in crops to induce useful mutations in rice, wheat, cotton, sweet pepper, tomato and sesame [50,52,53]. The powerful mutagens used to induce deletion mutations in plants are fast neutrons. Deletions might be as small as a few base pairs or as large as several kilobases.

5. FORWARD GENETICS

The approach of "Forward genetics" begins with the trait found in the organism and identified the respective gene using various approaches [54]. Transcriptome analysis, genetic transformation, and other techniques have been used to investigate the roles of genes, determine their significance in metabolic pathways, and investigate the regulatory mechanisms of genes at different developmental stages. The University of California, Davis' Tomato Genetics Resource Center (TGRC) has a library of over one thousand monogenic mutants that alter numerous aspects of plant growth. (<http://tgrc.uctavis.edu>). This collection of mutants includes naturally occurring and artificially induced mutations from various genetic background origins. Additionally, the TGRC has gathered close to 1,500 genetic stocks and on request, the seeds kept in the TGRC collection are made available for research. All of these mutant resources can help forward genetics' approach to identifying genes' functions. However, compared to reverse genetic approaches, a notable disadvantage of forward genetic approaches is that it takes them more time to correlate phenotype with genotype [55].

6. REVERSE GENETIC STRATEGIES

6.1 Reverse Genetics

The next step after the completion of whole genome sequencing of several organisms is to

give each gene a function. Reverse genetics is the creation or specific target recognition of mutations in a known gene sequence. As compared to forward genetics, which works from phenotype to genotype, reverse genetics works in a reverse way, from genotype to phenotype. Reverse genetics is the terminology alternative word for the functional genomics approach developed to determine the gene function by examining and comparing genetic variants in the target gene to phenotypic changes. One effective approach for identifying the activities of genes is gene silencing. In plants, several reverse genetics approaches are being developed, the majority of them rely on transformation and induced mutagenesis techniques. The development of efficient transformation and mutagenesis protocols enabled reverse genetics approaches [56]. Exogenous transposon and T-DNA insertional mutagenesis systems have been used in a reverse genetic approach known as insertional mutagenesis [57]. The transcriptional profiling is supported by the availability of a large EST collection in many plant species and successfully applied the sense and antisense silencing technology to identify the gene functions. The Flavr-savr® tomato, which has a delayed ripening process due to reduced expression of polygalacturonase in the cell wall, was the first transgenic plant to be introduced to the fresh market. It was created through antisense silencing and it might lengthen the shelf life and quality of storage of fruit, however, because of consumer concern about GMOs, it was a commercial not viable. Recently, tomato functional genomics has also shown progress with additional gene silencing technologies like RNA interference (RNAi) and virus-induced gene silencing (VIGS). Targeting Induced Local Lesions IN Genomes (TILLING), a unique reverse genetics technique developed by McCallum et al. [58], is based on the discovery of induced mutations in specific genes. Metabolomics is also played a crucial role to characterize and understanding plant diversity, biology, and improved quality breeding [59,60].

6.2 Gene Silencing

Antisense and sense RNAs can block the expression of a target gene through a process known as RNA interference (RNAi). It is based on the idea that double-stranded sequences can detect and destroy complementary sequences. One of the first systems where sense and antisense silencing were discovered to function efficiently was tomato fruit ripening [61,62].

Among the reverse genetics approaches, the RNAi-based silencing technique has become a potent tool for suppressing gene expression and examining the target gene-phenotype. An interesting reverse-genetics approach, virus-induced gene silencing (VIGS), has also been utilized successfully to identify the functions of genes. A virus vector technology called VIGS takes advantage of RNA resistance [63]. Post-transcriptional gene silencing (PTGS), a natural defense mechanism of plants, is used when plants were infected by a virus. The capability of delivering siRNAs to specific loci is the most significant limitation of these methods, which limits their high throughput screening. Additionally, it has the possibility of yielding unclear results and is unsuitable for mutant's isolation [64].

6.3 Targeting Induced Local Lesions in Genomes (TILLING)

Targeting Induced Local Lesions in Genomes merges both mutagenesis and high-throughput screening methodology for the detection of single-nucleotide changes in the genome [20]. TILLING is nontransgenic technology that is applied to a variety of crop species to enhance the desired traits [58]. It is a powerful tool for finding allelic series of point mutations, which is especially helpful for genes that are essential and alleles of sub-lethal. It has a number of advantages above other methods for finding point mutations. Other gel-based techniques, such as DGGE (denaturing gradient gel electrophoresis) and SSCP (single-strand conformation polymorphism) used to find single nucleotide alterations, are unable to identify the type or polymorphism location [64]. For the purpose of identifying mutations, the first TILLING method utilized a commercial denaturing HPLC (DHPLC) system [58]. But now they are different detecting methods were available for mutation detection, such as enzymatic mismatch cleavage-based techniques and Next generation sequencing (NGS). It is a high-throughput approach without the market and regulatory restrictions associated with transgenic crops.

6.4 Insertional Mutagenesis

Insertional mutagenesis approaches have been well-standardized for various plants. Transposons are genetic DNA elements with the ability to migrate across and can integrate into

different genomic regions. One disadvantage of this approach is the low frequency of mutations in the genome, which necessitates screening of a large number of mutagenized lines; furthermore, insertions in important genes are generally lethal.

6.4.1 T-DNA and transposon tagging

Genome sequences for many plant species, including Arabidopsis, rice, tomatoes, etc., were available in a database. Functional Genomics is defined as the "development and application of global (genome-wide or system-wide) experimental approaches to assess gene function by making use of the information and starting materials provided by structural genomics". T-DNA and Transposon have been the most commonly used tools for insertional mutagenesis in the reverse genetics approach. The T-DNA insertion is a random event and the insertion sequences remain stable over several generations. Insertional mutagenesis has been successfully applied to several plant species. In rice and Arabidopsis, approximately 1.5 insertions per genome were found which is to be a low copy number per genome [65]. Excision of the transposon from the disrupted gene in the presence of an Ac element allows for the simple confirmation of Ds transposon mutants. Using tools like TAIL-PCR, iPCR, genome walking, etc., it is simple to identify the flanking sequence of the gene in which T-DNA or transposon is inserted [66]. Because T-DNA insertions will remain stable over several generations, using T-DNA as the insertional mutagen is more practical than transposon tagging. However, the ability of transposons to move to adjacent loci and cause mutations in every member of a gene family makes them beneficial [67].

6.4.2 Gene trap and activation tagging

Insertional mutagenesis does have drawbacks, such as the inability to determine the function of redundant genes, and genes associated with gametophyte development. Further, it is also challenging to establish a link between the insertion and visible phenotypes. To address the limitations of the conventional insertional mutagenesis approach, methods like the Gene Trap and Activation Tagging were developed. Insertion of multimerized CaMV 35S enhancers was used in the activation tagging technique and its effect on transcriptional activation of nearby genes [68]. Genes that are overexpressed can result in a variety of traits that can be used to detect important genes, such as redundant gene

families and genes required for survival. One of the modified systems, the gene trap system, involves the fusion of reporter genes such as green fluorescent protein or glucuronidase (gus) with the tagged genes (GFP). Identification of novel genes based on their expression patterns is made easier by the fusion of tagged genes with reporter genes (gfp) [69]. In addition to disrupting normal gene function, the promoter-less reporter gene insertion also triggers the reporter gene's expression. The reporter gene activation provides information about the location and timing of the knockout gene expression, making the gene trap system useful for studying mutant phenotypes.

7. CONCLUSION

Mutagenesis is used to create raw material (mutants) for genetic improvement of crop plants as well as to investigate gene functions. Induced mutagenesis is a way to enhance genetic variability by exposing living cells/organisms to mutagens that may be chemical, physical, or biological agents. Mutagenesis research programs have generated a huge range of genetic variability and have played a substantial role in plant breeding programs all over the world. Genetic variations have been induced using ionizing radiation and chemical mutagens and mutation breeding has been successfully implemented to improve major crops such as wheat, rice, barley, cotton, peanuts, beans, etc. Historically, the mutagenesis approach was used in forward genetic screening, where better traits were selected and transferred into elite cultivars through breeding projects. The generation of mutagenized population in plants coupled with high-resolution reverse genetics and forward genetics screening methods allowed breeders to choose traits of interest and this has given a unique possibility to identify novel traits. Breeders used the genetic variation with heritable changes in various genetic backgrounds to develop new varieties.

COMPETING INTERESTS

Authors have declared that no competing interests exist.

REFERENCES

1. Pahalvi HN, Rafiya L, Rashid S, Nisar B, Kamili AN. Chemical fertilizers and their impact on soil health. In: Dar, G.H., Bhat, R.A., Mehmood, M.A., Hakeem, K.R. (eds) Microbiota and Biofertilizers, Springer, Cham. 2021;2.
2. Sharma A, Kumar V, Shahzad B, Tanveer M, Sidhu GPS, Handa N, Kohli SK, Yadav P, Bali AS, Parihar RD, Dar OI, Singh K, Jasrotia S, Bakshi P, Ramakrishnan M, Kumar S, Bhardwaj R, Thukral AK. Worldwide pesticide usage and its impacts on ecosystem. SN Appl. Sci. 2019; 1: 1446.
3. Chaudhary J, Deshmukh R, Sonah H. Mutagenesis approaches and their role in crop improvement. Plants. 2019;8:467.
4. Harlan JR. Crops and man. American Society of Agronomy and Crop Science Society of America, Madison; 1992.
5. Tracy WF. What is plant breeding?, in Sligh M, Lauffer L, eds., Proceedings of the Summit on Seeds and Breeds for 21st Century Agriculture. Washington, DC: Rural Advancement Foundation International-USA. 2004;23–30.
6. Adamu AK, Aliyu H. Morphological effects of sodium azide on tomato (*Lycopersicon esculentum* Mill). Sci World J. 2007;2:9–12.
7. Koornneef M. Classical mutagenesis in higher plants. Molecular Plant Biology. Gilmarin PM and Bowler C (eds.), Oxford, Oxford University Press. 2002;1–11.:
8. Neuffer MG, Coe EH, Wessler SR. Mutants of maize. (Cold Spring Harbor Laboratory Press: Cold Spring Harbor, NY, USA); 1997.
9. Razzaq A, Kaur P, Akhter N, Wani SH, Saleem F. Next-generation breeding strategies for climate-ready crops. Frontiers in Plant Science. 2021;12: 620420:1-27.
10. Futuyma DJ. Evolution (3rd Ed.) Sinauer; 2013. ISBN 978-1605351155.F.
11. Gregory TR. Insertion-deletion biases and the evolution of genome size. Gene. 2004;324:15–34.
12. Sallaud C, Gay C, Larmande P, Bès M, Piffanelli P, Piégu B, Droc G, Regad F, Bourgeois E, Meynard D, Périn C, Sabau X, Ghesquière A, Glaszmann JC, Delseny M, Guiderdoni E. High throughput T-DNA insertion mutagenesis in rice: a first step towards in silico reverse genetics. Plant J. 2004;39(3):450-64.
13. Panara F, Calderini O, Porceddu A. T-DNA Insertional Mutagenesis and Activation Tagging in *Medicago truncatula*. Methods Mol Biol. 2018;1822:83-105.

14. Upadhyaya NM, Zhu QH, Bhat RS. Transposon insertional mutagenesis in rice. *Methods Mol Biol.* 2011;678:147-77.
15. Griffiths AJF, Wessler SR, Carrol SB, John D. Introduction to genetics analysis (10th Edition) New York: W.H. Freeman and Company; 2012.
16. Durland J, Ahmadian-Moghadam H. Genetics, mutagenesis. Treasure Island (FL): StatPearls Publishing; 2022. PMID: 32809354.
17. Mba C, Afza R, Bado S, Mohan Jain S. Induced mutagenesis in plants using physical and chemical agents. *Plant Cell culture: Essential Methods.* 2010;20:111–130.
18. Till B, Reynolds S, Weil C, Springer N, Burtner C, Young K, Bowers E, Codomo C, Enns L, Odden A, Greene E, Comai L, Henikoff S. Discovery of induced point mutations in maize genes by TILLING. *BMC Plant Biol.* 2004;4:1-8.
19. Slade AJ, Fuerstenberg SI, Loeffler D, Steine MN, Facciotti D. A reverse genetic, nontransgenic approach to wheat crop improvement by TILLING. *Nature Biotechnology.* 2005;23(1):75–81.
20. Till B, Cooper J, Tai T, Colowit P, Greene E, Henikoff S, Comai L. Discovery of chemically induced mutations in rice by TILLING. *BMC Plant Biol.* 2007;7:1-12.
21. Triques K, Sturbois B, Gallais S, Dalmais M, Chauvin S, Clepet C, Aubourg S, Rameau C, Caboche M, Bendahmane A. Characterization of *Arabidopsis thaliana* mismatch specific endonucleases: application to mutation discovery by TILLING in pea. *Plant J.* 2007;51(6):1116-25.
22. Cooper JL, Till BJ, Laport RG, Darlow MC, Kleffner JM, Jamai A, El-Mellouki T, Liu S, Ritchie R, Nielsen N, Bilyeu KD, Meksem K, Comai L, Henikoff S. TILLING to detect induced mutations in soybean. *BMC Plant Biol.* 2008;24;8:9.
23. Talame V, Bovina R, Sanguineti MC, Tuberosa R, Lundqvist U, Salvi S. TILLMore, a resource for the discovery of chemically induced mutants in barley. *Plant Biotechnology Journal.* 2008;6(5): 477–485.
24. Wang N, Wang Y, Tian F, King GJ, Zhang C, Long Y, Shi L, Meng J. A functional genomics resource for *Brassica napus*: development of an EMS mutagenized population and discovery of FAE1 point mutations by TILLING. *New Phytol.* 2008;180(4):751-65.
25. Xin Z, Wang ML, Barkley NA, Burow G, Franks C, Pederson G, Burke J. Applying genotyping (TILLING) and phenotyping analyses to elucidate gene function in a chemically induced sorghum mutant population. *BMC Plant Biol.* 2008;14;8:103.
26. Gady ALF, Hermans FWK, Van de Wal MHB, Van Loo EN, Visser RGF, Bachem CWB. Implementation of two high throughput techniques in a novel application: detecting point mutations in large EMS mutated plant populations. *BMC Plant Methods.* 2009;5:13.
27. Himelblau E, Gilchrist EJ, Buono K, Bizzell C, Mentzer L, Vogelzang R, Osborn T, Amasino RM, Parkin IA, Haughn GW. Forward and reverse genetics of rapid-cycling *Brassica oleracea*. *Theor Appl Genet.* 2009;118(5):953-61.
28. Dalmais M, Schmidt J, Le Signor C, Moussy F, Burstin J, Savoie V, Aubert G, Brunaud V, de Oliveira Y, Guichard C, Thompson R, Bendahmane A. UTILLdb, a *Pisum sativum* in silico forward and reverse genetics tool. *Genome Biol.* 2008;9(2):R43.
29. Chawade A, Sikora P, Bräutigam M, Larsson M, Vivekanand V, Nakash MA, Chen T, Olsson O. Development and characterization of an oat TILLING-population and identification of mutations in lignin and beta-glucan biosynthesis genes. *BMC Plant Biol.* 2010;12;10:86.
30. Stephenson P, Baker D, Girin T, Perez A, Amoah S, King GJ, Østergaard L. A rich TILLING resource for studying gene function in *Brassica rapa*. *BMC Plant Biol.* 2010;9:10:62.
31. Sabetta W, Alba V, Blanco V, Montemurro C. SunTILL: a TILLING resource for gene function analysis in sunflower. *Plant Methods.* 2011;7(1):20.
32. Jander G, Baerson SR, Hudak JA, Gonzalez KA, Gruys KJ, Last RL. Ethylmethanesulfonate saturation mutagenesis in *Arabidopsis* to determine frequency of herbicide resistance. *Plant Physiol.* 2003;131(1):139-46.
33. Nasri F, Zakizadeh H, Vafaei Y, Mozafari AA. In vitro mutagenesis of *Chrysanthemum morifolium* cultivars using ethylmethanesulphonate (EMS) and mutation assessment by ISSR and IRAP markers. *Plant Cell Tiss Organ Cult.* 2022;149:657–673.

34. Siddique MI, Back S, Lee JH, Jo J, Jang S, Han K, Venkatesh J, Kwon JK, Jo YD, Kang BC. Development and Characterization of an Ethyl Methane Sulfonate (EMS) Induced Mutant Population in *Capsicum annuum* L. Plants (Basel). 2020;9(3):396.
35. Perry JA, Wang TL, Welham TJ, Gardner S, Pike JM, Yoshida S, Parniske MA. TILLING reverse genetics tool and a web-accessible collection of mutants of the legume *Lotus japonicus*. Plant Physiol. 2003;131(3):866-71.
36. González M, Xu M, Esteras C, Roig C, Monforte AJ, Troadec C, Pujol M, Nuez F, Bendahmane A, Garcia-Mas J, Picó B. Towards a TILLING platform for functional genomics in Piel de Sapo melons. BMC Res Notes. 2011;11(4):289.
37. Opoku Gyamfi M, Eleblu JSY, Sarfoa LG, Asante IK, Opoku-Agyemang F, Danquah EY. Induced variations of ethyl methane sulfonate mutagenized cowpea (*Vigna unguiculata* L. walp) plants. Front Plant Sci. 2022;5(13):952247.
38. Heffron LM, Korban SS. Mutagenic responses to ethyl methanesulfonate and phenotypic characterization of an M₁ generation of snapdragon, *Antirrhinum majus*. Euphytica. 2022;218: 76.
39. Goyal S, Wani MR, Raina A, Laskar RA, Khan S. Phenotypic diversity in mutagenized population of urdbean (*Vigna mungo* (L.) Hepper). Heliyon. 2021;12(7(5):e06356.
40. Karaman K, Kizil S, Başak M, Uzun B, Yol E. Development of EMS-induced Mutagenized Groundnut Population and Discovery of Point Mutations in the ahFAD2 and Ara h 1 Genes by TILLING. J Oleo Sci. 2021;70(11):1631-1640.
41. Latado RR, Adames HA, Neto TA. *In vitro* mutation of chrysanthemum (*Dendranthema grandiflora* Tzvelev) with ethylmethanesulphonate (EMS) in Immature Floral Pedicels. Plant Cell, Tissue and Organ Culture. 2004;77:103–106.
42. Muller H. The production of mutations by X-rays. PNAS of the USA. 1928;14:714.
43. Stadler LJ. Mutations in barley induced by X-rays and radium. Science. 1928;68,186–187.
44. Stadler LJ. Some genetic effects of X-rays in plants. J. Heredity. 1930;21:2–19.
45. Amri-Tiliouine W, Jankowicz-Cieslak J, Till BJ, Laouar M. Generation of mutant plants by gamma ray exposure and development of low-cost TILLING population in chickpea (*Cicer arietinum* L.). Methods Mol Biol. 2022;2484:143-159.
46. Komura S, Jinno H, Sonoda T, Oono Y, Handa H, Takumi S, Yoshida K, Kobayashi F. Genome sequencing-based coverage analyses facilitate high-resolution detection of deletions linked to phenotypes of gamma-irradiated wheat mutants. BMC Genomics. 2022;9:23(1):111.
47. Elia PPJ, Nambi Rajesh, Jogi Gurucharan Padma EC, Surendranatha Reddy, Pinjari Osman Basha. Generation of gamma irradiated mutagenized population in *Solanum lycopersicum* CV. Arka Vikas. Current Trends in Biotechnology and Pharmacy. 2020;14(3):271-278.
48. Chen Y, Chen R. Physical Mutagenesis in *Medicago truncatula* Using Fast Neutron Bombardment (FNB) for Symbiosis and Developmental Biology Studies. Methods Mol Biol. 2018;1822:61-69.
49. Bose JL. Chemical and UV mutagenesis. Methods Mol Biol. 2016;1373:111-5.
50. Mei, M, Deng H, Lu Y, Zhuang C, Liu Z, Qiu Q, Qiu Y, Yang TC. Mutagenic effects of heavy ion radiation in plants. Adv. Space Res. 1994;14:363–372.
51. Wu, L, Yu Z. Radiobiological effects of a low-energy ion beam on wheat. Radiat. Environ. Biophys. 2001;40:53–57.
52. Ren W, Zhang Y, Deng B, Guo H, Cheng L, Liu Y. Effect of space flight factors on alfalfa seeds. Afr. J. Biotechnol. 2010;9:7273–7279.
53. Liu L, Zhao L, Guo H. Current status and perspective outlook of space induced mutation breeding in crop plants. Rev. China Agric. Sci. Technol. 2007;9:26– 29.
54. Ali Khan A, Raess M, de Angelis MH. Moving forward with forward genetics: A summary of the INFRAFRONTIER Forward Genetics Panel Discussion. F1000Res. 2021;8:10:456.
55. Gilchrist E, Haughn G. Reverse genetics techniques: engineering loss and gain of gene function in plants. Briefings in Functional Genomics. 2010;9(2):March:103–110.
56. Rajesh N, Kumar JS, Prashanth JE, Basha PO. An establishment of efficient *Agrobacterium* mediated transformation in tomato (*Solanum lycopersicum*). IJRSR. 2016;7(1): 8583-8591.
57. Meissner R, Chague V, Zhu Q, Emmanuel E, Elkind Y, Levy AA. Technical advance:

- a high throughput system for transposon tagging and promoter trapping in tomato. *The Plant Journal*. 2000;22(3):265–274.
58. McCallum CM, Comai L, Greene EA, Henikoff S. Targeted screening for induced mutations. *Nature Biotechnology*. 2000;18: 455-457.
59. Fernie AR, Schauer N. Metabolomics-assisted breeding: a viable option for crop improvement? *Trends in Genetics*. 2009;25(1):39-48.
60. De Vos RCH, Hall RD, Moing A. Metabolomics of a model fruit: Tomato. *Annual Plant Reviews*. 2011;109-155.
61. Smith CJ, Watson CF, Bird CR, Ray J, Schuch W, Grierson D. Expression of a truncated tomato polygalacturonase gene inhibits expression of the endogenous gene in transgenic plants. *Mol Gen Genet*. 1990;224:477-481.
62. Gray J, Picton S, Shabbeer J, Schuch W, Grierson D. Molecular biology of fruit ripening and its manipulation with antisense genes. *Plant Molecular Biology*. 1992;19(1): 69–87.
63. Lu R, Martin-Hernandez AM, Peart JR, Malcuit I, Baulcombe DC. Virus-induced gene silencing in plants. *Methods*. 2003;30:296–303.
64. Gilchrist EJ, Haughn GW. TILLING without a plough: a new method with applications for reverse genetics. *Curr Opin Plant Biol*. 2005;8:211–215.
65. Jeon JS, Lee S, Jung KH, Jun SH, Jeong DH, Lee J, Kim C, Jang S, Yang K, Nam J, An K, Han MJ, Sung RJ, Choi HS, Yu JH, Choi JH, Cho SY, Cha SS, Kim SI, An G. T-DNA insertional mutagenesis for functional genomics in rice. *Plant J*. 2000;22:561–570.
66. Liu YG, Whittier R. Thermal asymmetric interlaced PCR: automatable amplification and sequencing of insert end fragments from P1 and YAC clones for chromosome walking. *Genomics*. 1995;25:674–681.
67. Krysan PJ, Young JC, Sussman MR. T-DNA as an insertional mutagen in *Arabidopsis*. *Plant Cell*. 1999;11:2283-2290.
68. Suzuki Y, Uemura S, Saito Y, Murofushi N, Schmitz G, Theres K, Yamaguchi I. A novel transposon tagging element for obtaining gain-of-function mutants based on a self-stabilizing *Ac* derivative. *Plant Mol Biol*. 2001;45:123–131.
69. Springer PS. Gene traps: tools for plant development and genomics. *Plant Cell*. 2000;12:1007–1020.